Exercise on Matplotlib

1. Create two lists of length 10 with numeric values of your choice and plot them. Plot them using pyplot.
2. Using list comprehension, create a list, where each element is the sum of the current integer and its previous integer. Use range(20) as x-axis and plot the generated list as y-axis.
    (a) Set the title for the plot.
    (b) Set x label and y label of your choice.
    (c) Set line width to 3.
    (d) Experiment with at least 3 different combinations of color, marker, size, and font, and choose your best one.
3. Using NumPy, generate two arrays of random values: one in the range 5 to 10 (for x-axis) and another in the range 1 to 5 (for y-axis). Plot these values using a scatter plot in Matplotlib.
    (a) Set x label, y label, plot title.
    (b) Change Marker shape, marker size and color
    (c) change transparency for 3 different "alpha".
4. Create a list of 30 runs (values between 0 to 100 generated randomly) representing scores by Virat Kohli in 30 matches. Create histogram,
    (a) With default bin edges
    (b) With Bin edges of your choice
5. Create an empty figure and axes(one) using subplots() function.
6. Create an empty figure with size of your choice having 2 rows and one column.
    (a) Try 3 different heights and widths.
7. Plot a Sine wave using subplots with only one axes.

(a) Set axes title, X label, Y label.

(b) Create legend for the plot.

(c) Place the legend in 4 different locations.

8. Create a figure containing 2x2 subplots. Plot as mentioned in Question number (7) in the bottom right axes. Others may be kept empty.

   (a) In Top left axes.

   (b) Create figure title as "Figure Title” with appropriate font size. And plot in bottom left axes with axes title as "Axes plot is here".

9. Create a figure containing 2x2 subplots. Plot in all four axes with Figure title and axes titles of your Choice. Plotting variables are of your choice. Use legends, texts.

10. Create a 2x1 subplot. Share "X" axis. "Y" variable for bottom axes is Sin(2* Pi*t/2). For top axes, Sin(2* Pi*t/4). Set titles, labels, legends etc (all that you have learned). (Hint. Use NumPy Linspace for t.